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Research Statement

Research interests: Applied Microeconomics, Production Econometrics, and Industrial Organization.

I am an applied microeconomist studying productivity and technology adoption, with expertise in production econometrics. My research examines how technological improvements translate into firm performance and why the gains differ across firms. My job market paper separates digitalization's relationships with labor efficiency, broader productivity, and markups in Spanish manufacturing. My banking research examines a complementary puzzle: technological progress can reduce costs even as realized productivity declines. Together, these papers show why evaluating technology requires distinguishing improvements in production possibilities from changes in firms' actual performance. My mediation research investigates whether digitalization's relationship with output operates through additional labor or greater efficiency. My work on heterogeneous treatment effects provides guidance on choosing methods to study differences in who benefits.

These projects motivate my independent agenda on reliable productivity measurement and the diffusion of technology's benefits. In my solo regularization project, I examine whether methods that predict output accurately also recover reliable input elasticities, returns to scale, and markups. My planned solo AI project extends the adoption framework to productivity spillovers through geographic proximity and economic linkages. This agenda connects the measurement of gains within firms to understanding how those gains extend beyond adopters.

Job Market Paper: Digitalization, Factor-Biased Productivity, and Market Power

Higher revenues do not reveal how digital investments change production. Technology may help workers produce more with the same hours or improve efficiency across inputs. It may also enable firms to charge higher prices relative to costs. Distinguishing these channels helps explain what adoption achieves. It also informs the evaluation of policies that subsidize adoption. My job market paper, **“Digitalization, Factor-Biased Productivity, and Market Power”** (with Subal C. Kumbhakar), examines these channels within a common framework. The application uses Spanish manufacturing firms over 2000–2019.

The main finding is that digitalization is associated with modest improvements in productive efficiency. The corresponding change in markups is much smaller. The digitalization index ranges from zero to one. In the model allowing firms to have market power, a 0.10-point increase is associated with approximately **0.55 percent higher labor-augmenting productivity** in the following period. The corresponding estimates are **0.43 percent for Hicks-neutral productivity** and **0.07 percent for markups**. Labor-augmenting productivity captures the efficiency of labor hours. Hicks-neutral productivity captures efficiency across the input bundle. In the price-taking benchmark, the corresponding productivity estimates are 0.37 and 0.62 percent, respectively. These estimates describe conditional relationships under the model's assumptions.

The gains also differ across firms. The labor-efficiency association is positive and statistically significant for small firms under both market environments. This association is significantly larger than that for large firms, whose estimate is imprecise. Among small firms, IT technologies, digital human capital, and automation each have positive labor-efficiency associations in the model with market power. The

markup response is significantly smaller in industries with high digital intensity. These findings motivate evaluating adoption programs by the productivity channels, technologies, and firms involved.

I use the Encuesta Sobre Estrategias Empresariales (ESEE), a Spanish manufacturing dataset with detailed information on production and digitalization. I construct a digitalization index from thirteen variables. These cover IT, digital human capital, automation, and interaction with customers and suppliers. I develop a sequential identification approach that distinguishes labor efficiency, broader productivity, and markup dynamics under imperfect competition. The approach identifies technology parameters before recovering markups from input expenditure shares. It does not require firm-level input-price data. This sequence addresses the difficulty that the same shares reflect both technology and pricing behavior.

Productivity and Technological Change in Banking

My banking publication examines a related puzzle: why can technological progress and consolidation coexist with declining productivity? The paper, **“The Scale–Technology Paradox: Technological Progress and Productivity Decline in Nigerian Banking,”** studies seventeen Nigerian commercial banks over 2000–2023. It is coauthored with Subal C. Kumbhakar and published in *Economic Modelling* (2026). The analysis accounts for banks’ multiple outputs and endogenous input choices. It distinguishes changes in production possibilities from changes in realized productivity.

I estimate an average annual productivity decline of **3.5 percent**. The declines are steeper among larger and internationally licensed banks. Yet technical change reduces costs. Estimated returns to scale average 1.46, indicating substantial potential gains from expansion. Lending is the principal source of cost pressure. The economic lesson is that modernization and greater scale need not be sufficient to improve performance. The findings motivate attention to lending efficiency, credit information, collateral enforcement, and bad-loan resolution. This project extends my productivity research to financial intermediation. It highlights the importance of understanding how firms use improved technologies.

Causal Mechanisms and Heterogeneous Effects

Understanding the returns to adoption also requires identifying the channels through which output changes. In **“Mediation Analysis in Production Functions with Unobserved Productivity”** (with Subal C. Kumbhakar), I study digitalization’s relationship with output. The paper asks how much of this relationship operates through changes in labor hours. More productive firms may both adopt technology and adjust employment. This makes it difficult to distinguish a labor-mediated effect from differences in underlying productivity. The paper develops a framework that controls for unobserved productivity in both the production and labor-hours equations. Simulations show that omitting this control from the labor equation can substantially overstate the indirect effect. An application to Spanish manufacturing illustrates the economic importance of this issue. It distinguishes producing more through additional labor from producing more efficiently.

My published paper, **“Causal Forests versus Penalized Splines for Heterogeneous Treatment Effects,”** examines how treatment-effect variation should guide empirical method choice. It is coauthored with Ivan Korolev and Nancy Dhameja and published in *Economics Letters* (2026). In simulations with randomized assignment, forests perform best when effects change sharply. Penalized splines generally perform better with smooth heterogeneity. They also perform better in designs combining continuous, binary, and group-level covariates. The findings provide practical guidance for researchers studying who benefits from an intervention. They inform my broader interest in matching empirical methods to the economic quantities researchers want to learn.

Solo Project: Regularization in Translog Production Models

In my solo project, **“Regularization in Translog Production Models,”** I ask whether better output prediction also improves the economic conclusions drawn from production models. Flexible models allow the contribution of an input to vary with a firm’s input mix. Their squared and interaction terms can, however, be highly correlated. Consequently, a model may predict output accurately while producing unstable estimates of the contribution of labor or capital. This instability matters because input elasticities underpin conclusions about returns to scale and production-based markups.

I am developing a comparison of Lasso, Ridge, Elastic Net, principal component regression, and partial least squares. The comparison evaluates how well these methods recover economically meaningful quantities. The planned simulations will vary input correlations, sample size, and the underlying production relationships. I will compare the bias and variability of input elasticities. I will also evaluate the accuracy of returns-to-scale estimates and the frequency of economically implausible estimates, such as negative marginal products. A further assessment will trace how elasticity errors carry through to inferred markups. The central question is whether choosing regularization for predictive accuracy yields different conclusions from choosing it for elasticity recovery.

The project’s goal is to show when regularization stabilizes economically relevant estimates and when it conceals distortions. I plan to assess its use alongside methods that address endogenous input choices. Regularization alone does not resolve input endogeneity. This project develops an independent strand of my agenda focused on the recovery of economically meaningful quantities. I evaluate methods by their ability to support economic interpretation and policy analysis.

Research Beyond the Dissertation

My next solo project, **“Artificial Intelligence and Spatial Factor-Biased Productivity,”** asks who benefits from AI beyond the adopting firm. It extends the adoption framework to examine productivity spillovers. AI may improve the efficiency of particular inputs within firms. Knowledge flows, worker mobility, and supply-chain relationships may transmit gains to other firms. I plan to compare exposure through geographic proximity and economic linkages. I will also assess whether the framework can distinguish spillovers from common shocks. The empirical design will require information on the AI applications firms use and the production activities they affect.

Beyond the dissertation, I plan to connect heterogeneous treatment-effect methods with production models. This work will examine which firms benefit most from adoption. It will also study which complementary investments help explain differences in those benefits. A priority is to accommodate endogenous adoption and input choices while retaining interpretable productivity measures. My regularization project will support this agenda through criteria for selecting methods that reliably recover elasticities and related economic outcomes. Together, these projects will examine how technology improves productive efficiency, why its benefits are uneven, and how gains extend across firms and industries.

References

Publications

- [1] “Causal Forests versus Penalized Splines for Heterogeneous Treatment Effects,” with Ivan Korolev and Nancy Dhameja. *Economics Letters*, 268, 113205, 2026. doi:10.1016/j.econlet.2026.113205.

- [2] “The Scale–Technology Paradox: Technological Progress and Productivity Decline in Nigerian Banking,” with Subal C. Kumbhakar. *Economic Modelling*, 107859, 2026.
doi:10.1016/j.econmod.2026.107859. Replication code.

Job Market Paper

- [3] “Digitalization, Factor-Biased Productivity, and Market Power,” with Subal C. Kumbhakar. Job market paper, September 2026.

Working Papers

- [4] “Mediation Analysis in Production Functions with Unobserved Productivity,” with Subal C. Kumbhakar.

Work in Progress

- [5] “Regularization in Translog Production Models.” Solo-authored.
- [6] “Artificial Intelligence and Spatial Factor-Biased Productivity.” Solo-authored.